Charge to Mass Ratio of the Electron

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1 Introduction and Background

In 1897, J.J. Thompson performed an experiment that produced beams of electrons from a thermally excited metal and deflected them using a magnetic field, causing them to follow circular paths. In doing so, he justified that these emissions are composed of constituents with both charge and mass: experiencing an interaction from a magnetic field while moving supported that these beams' constituents had nonzero charge, and traveling in a circular arc with nonzero radius caused by a force from a magnetic field supported that that they had nonzero mass. Thompson continued, obtaining a value of the charge to mass ratio of the beam constituents that was consistent across different heated metals. This result validated the notion of an electron as a fundamental particle that is the same across the materials it is found in [4,5]. In this experiment deriving from Thompson's original procedure, the ratio of the electron's charge to mass $\frac{e}{m}$ is determined by measuring the radius of the circular path of the electrons in a magnetic field produced by a pair of Helmholtz coils.

As part of this experiment, electrons initially at potential $V_0 = 0$ are accelerated by an anode with potential V_a and freely pass through a space with a uniform magnetic field $\vec{B}$. The Lorentz force on an electron with charge $-e$ moving with velocity $\vec{v}$ through this magnetic field $\vec{B}$ and negligible electric field is given by [2, 5]:

$$\vec{F} = -e\vec{v} \times \vec{B} \tag{1}$$

Because this force is orthogonal to the electron's velocity and to the field direction, the electrons travel in a circular arc in the plane normal to the field direction. Using conservation of energy principles and the uniform circular motion radial acceleration constraint, the radius of this arc r, the anode voltage V_a, the magnetic field strength B within the coils, the elementary charge e, and the electron mass m are related by [4, 5]:

$$2V_a = \left(\frac{e}{m}\right) B^2 r^2 \tag{2}$$

A pair of Helmholtz coils is used to generate the uniform magnetic field the electrons experience. The magnetic field strength B within the coils causing the circular acceleration can be expressed as a function of the current I through the coils, the number of turns N, the diameter D, and the vacuum permeability μ_0 [1]:

$$B = \frac{16\mu_0 N I}{\sqrt{125} D} \tag{3}$$

2 Methodology

An experimental setup is shown in Figure 1. A pair of Helmholtz coils encloses a coordinate grid positioned next to an anode, in which a slit is cut and through which electrons freed from a filament can pass [4, 5]. The coils are supplied a current via a DC power supply, at which point a current through the coils can be measured [1, 5]. The anode is raised to a voltage V_a on the order of kilovolts, while the filament is held at a potential on the order of volts. This causes freed electrons to flow into the space in front of the coordinate grid and form a circular arc according to Equation 1. A pair of electrodes above and below the coordinate grid, raised to the same large potential as the anode, ensure the electric field influence on the electron path is negligible [2]. The space within the coils is evacuated to a vacuum to minimize electron scattering by air molecules and to ensure the only acceleration on the electrons is only due to the magnetic field.

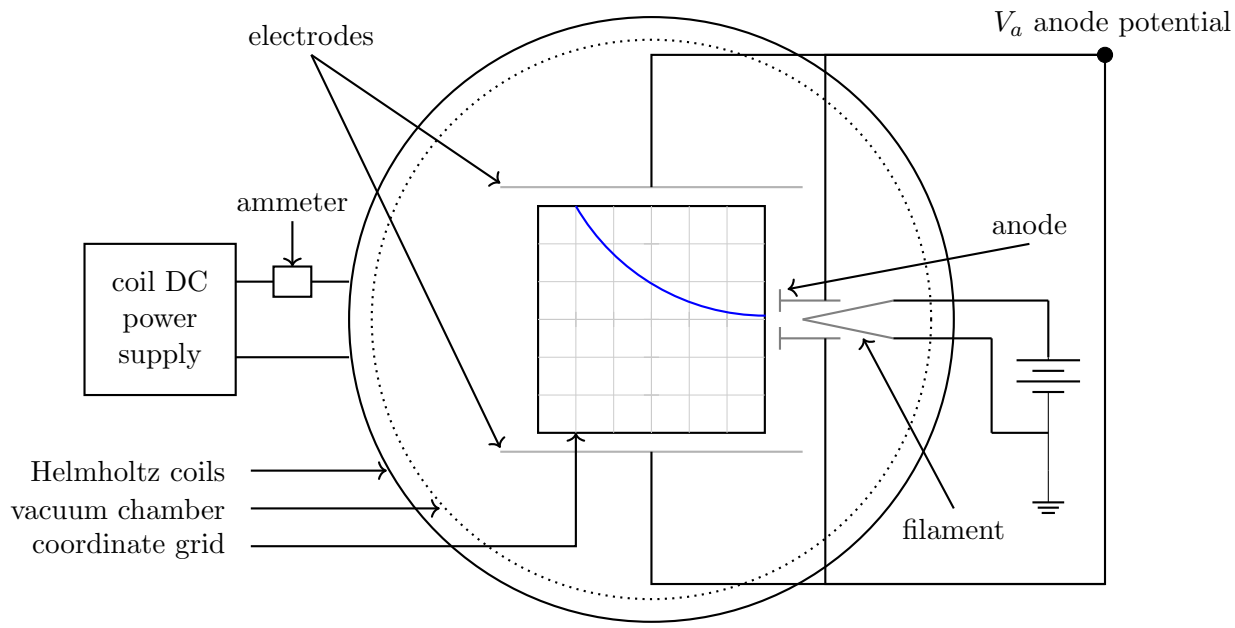


Figure 1: The experimental setup used to deflect emitted electrons. A pair of Helmholtz coils produce a uniform magnetic field [1] through a coordinate grid, over which electrons freed from a thermally-excited filament can pass—indicating their path. The electrons are given an initial velocity by accelerating toward an anode, which is raised to a voltage V_a [2, 5]. Electrodes at potential V_a above and below the coordinate grid prevent electric field influences on the electron path by rendering potential gradients in the region negligible [2]. The space in which electrons travel is enclosed in a vacuum chamber. Current is supplied to the coils via a DC power supply, another voltage source heats the filament, and a high-voltage source provides the anode and electrode voltages.

Anode voltages of $V_a = 1000$, 1500, 2000, 2500, and 3000 volts are produced and correspond to electron streams of various speeds–and thus to Lorentz forces of different magnitudes. Once a stream of electrons is established, the magnetic field of the coils is adjusted to produce an arc of approximately the same 3.3 cm radius as the arcs in the sibling trials. This is done by targeting a particular reference point on the path per trial. The coil current I accomplishing this is recorded and y-coordinates on the arc corresponding to the x-coordinates $[x_i]_i = [2.0, 4.0, 6.0, 8.0, 10.0]$ cm are measured using the coordinate grid.

It is possible that the anode slit does not align perfectly with the x-axis of the coordinate grid or that other magnetic fields influence the electron path. Two measures address these issues: correction using zero-field offsets and reversal of the magnetic field. The first measure involves the following: y-coordinates of the electron path without an active magnetic field are measured and subtracted from the y-coordinate readings of the electron path with an active magnetic field. The second measure is as follows: Upon collecting arc y-coordinates across all voltages, the magnetic field is reversed such that the electron path's y-coordinate is inverted. For each anode voltage, the current is brought to the same magnitude as in the corresponding non-inverted trial and coordinates of points on the arc are again measured, corresponding to the same x-coordinates. For each voltage, the entire data-collection process yields two lists of y-coordinates: an original list $[\hat{y}_i]_i$ and the inverted-field list $[\hat{y}'_i]_i$.

3 Results

The y-coordinate magnitudes obtained from each trial and its inverted-field counterpart are averaged, producing a single list of y-coordinates: $\left[\frac{|\hat{y}_i|+|\hat{y}'_i|}{2}\right]_i = [y_i]_i$. Fitting a circle's equation to the points $[(x_i, y_i)]_i$ yields one path radius r per anode voltage. Associated path radii, anode voltages, and coil currents are shown in Table 1 and an example of a circle fit to the coordinates of an electron path is displayed in Figure 2.

For each anode voltage in Table 1, a value of the electron charge to mass ratio $\frac{e}{m}$ is calculated using a rearranged form of Equation 2, with values of B obtained from Equation 3 using a coil turn number of 320 and a coil diameter of (15.5 ± 0.5) cm. These values are also shown in Table 1. Taking the mean of these pointwise values and performing a standard deviation of mean analysis produces an experimental value of $\frac{e}{m} = (1.68 \pm 0.05) \times 10^{11}$ C/kg. The accepted CODATA value for $\frac{e}{m}$ lies within 2σ of this experimental value [3].

$\frac{e}{m}$ data points, the experimental value, and the accepted CODATA value are displayed in Figure 3. Inspection of this plot reveals that the trials corresponding to anode voltages 2500 and 3000 volts produced values considerably lower than the accepted value.

Anode Voltage (V) ($\pm$1 V)	r (m)	I (A) ($\pm 0.0001A$)	e/m (10^{11} C/kg)
1000	0.33 ± 0.04	0.0850	1.8 ± 0.5
1500	0.327 ± 0.011	0.1052	1.72 ± 0.16
2000	0.318 ± 0.018	0.1238	1.75 ± 0.23
2500	0.34 ± 0.04	0.1393	1.5 ± 0.3
3000	0.326 ± 0.026	0.1540	1.62 ± 0.28

Table 1: Measured values of anode voltage V_a, electron path radius r, coil current I, and calculated electron charge-to-mass ratio (e/m) for each trial.

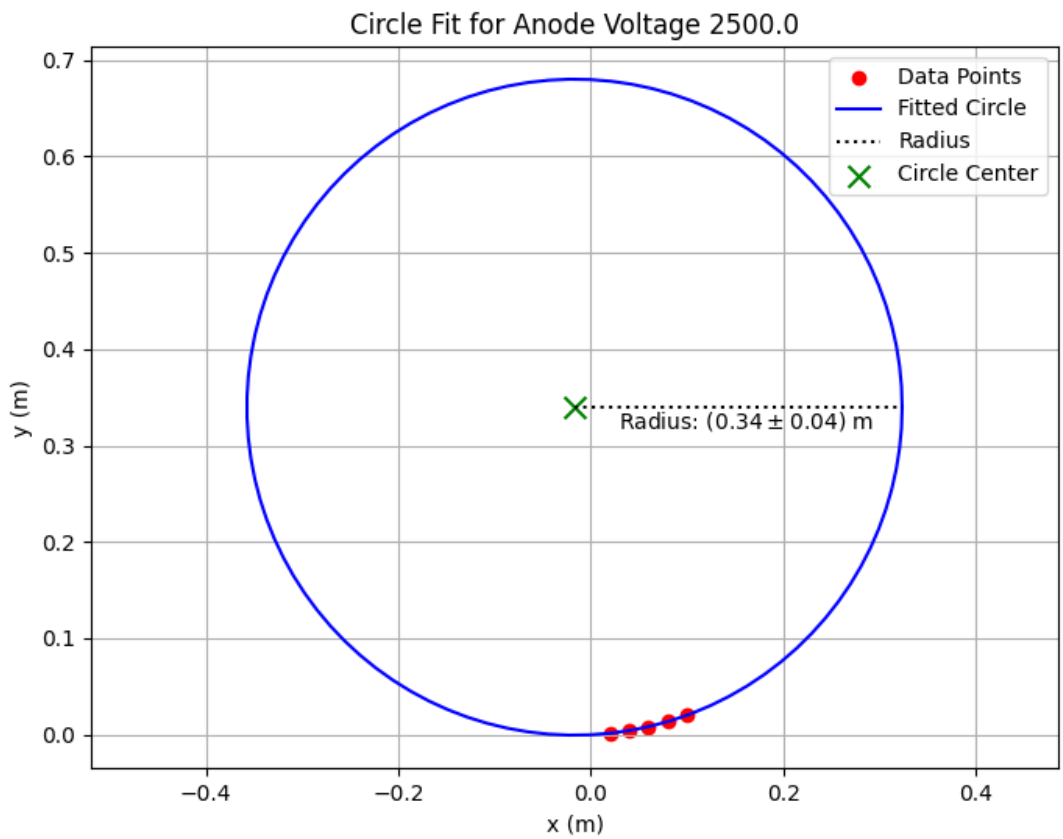


Figure 2: An example of a circle fit to points measured from an electron beam's path using least squares methods, with a radius obtained as a result of fitting.

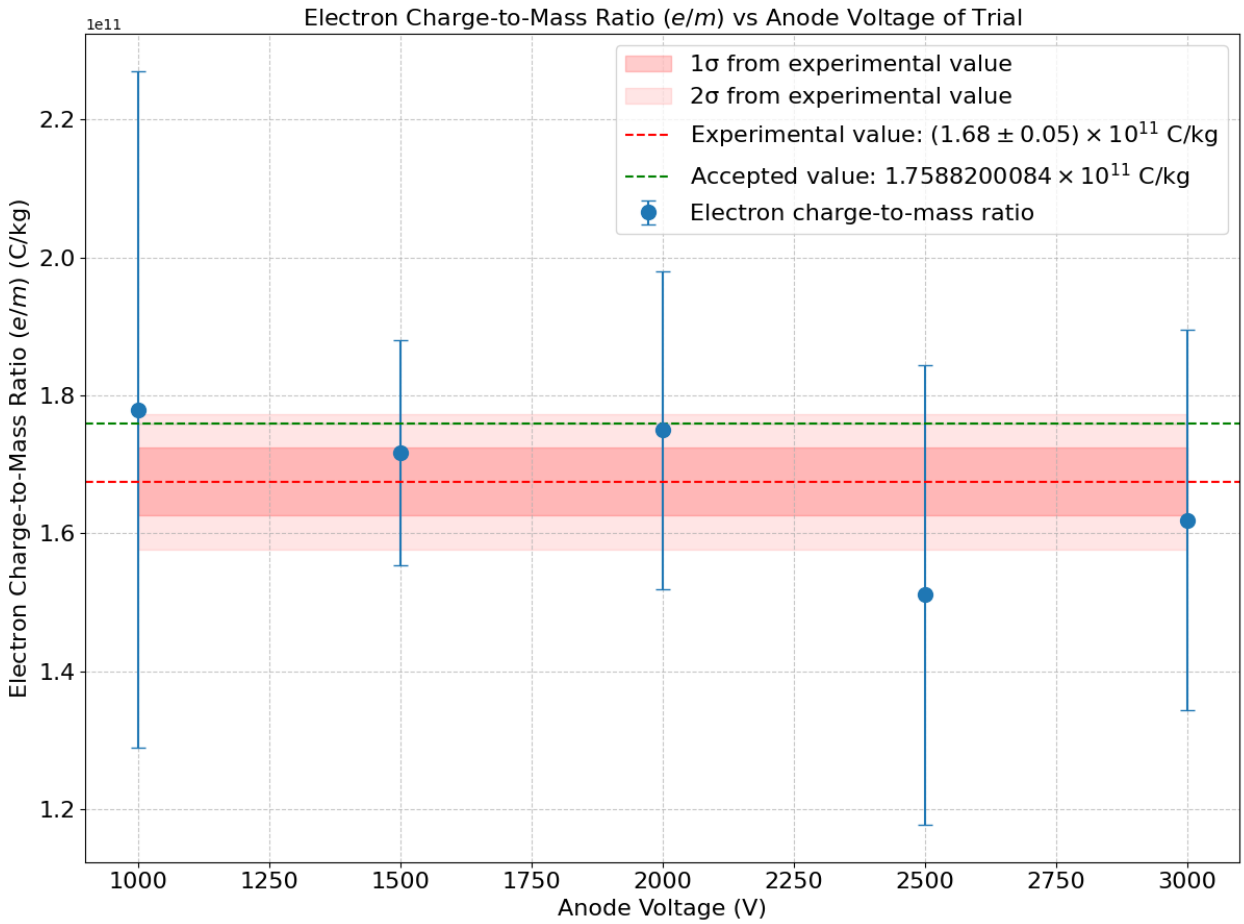


Figure 3: A plot indicating measured values of the electron charge to mass ratio derived from the values of $\frac{e}{m}$ in Table 1. 1σ and 2σ intervals around the dashed red experimental value are shaded in red, and the accepted CODATA value [3] is shown in green. Note that the points corresponding to anode voltages 2500 and 3000 volts are considerably lower than the accepted value.

4 Conclusion and Discussion

This experiment finds a value of the electron charge to mass ratio, $(1.68 \pm 0.05) \times 10^{11}$ C/kg, that is in agreement with the accepted CODATA value to within 2σ. It is possible that the electrodes above and below the coordinate grid were not perfectly aligned or that there are imperfections in the electrode materials, resulting in asymmetric electric field effects between the original trials and the field-flipped trials. Uniform magnetic fields are also a physical ideal; the coils' field might have had non-uniformities from one point on the coordinate grid to another [1, 5]. Any of these factors could have created a path that is not entirely circular, invalidating equation 2, which assumes uniform circular motion.

In particlar, it was noticed that paths with inverted fields tended to have larger y-coordinates than paths with non-inverted fields. A negligible zero-field deflection was observed, suggesting that this asymmetry could be due to the aforementioned non-uniformities in the magnetic field or in the electrodes' effects.

Qualitatively, it is noted that the electron beams follow roughly circular arcs through the magnetic field in agreement with the theoretical acceleration they would experience from the Lorentz force described by Equation 1: for a leftward-pointing initial electron velocity $\vec{v}_0$ and a magnetic field $\vec{B}$ pointing into the coordinate grid, the electrons experience a force antiparallel with the cross product $\vec{v}_0 \times \vec{B}$—or downward along the coordinate grid.

References

[1] B. Barman and A. Petrou. Measuring the magnetization of a permanent magnet. *American Journal of Physics*, 87(4):275–278, 2019.

[2] David Jeffrey Griffiths and Cambridge University Press. *Introduction to Electrodynamics*. Cambridge University Press, Cambridge I Pozostałe, 4 edition, 2018.

[3] Peter J. Mohr, David B. Newell, Barry N. Taylor, and Eite Tiesinga. CODATA Recommended Values of the Fundamental Physical Constants: 2022. *Reviews of Modern Physics*, 97(2):025002, 2025.

[4] George E. Smith. J. J. Thomson and The Electron: 1897–1899 An Introduction. *The Chemical Educator*, 2(6):1–45, 1997. Publisher: Springer-Verlag New York, Inc.

[5] Hugh D Young and Roger A Freedman. *University Physics: with Modern Physics*. Harlow Pearson Education Limited, 15 edition, 2020.

Analysis, Question 5

$B = \frac{16 \mu_0 N I}{\sqrt{125}\, D}$. Using the fractional propagation methods from Taylor, page 61:

$$\frac{\delta B}{|B|} = \left(\left(\frac{\delta I}{|I|}\right)^2 + \left(\frac{\delta D}{|D|}\right)^2 \right)^{1/2}$$

Using the rule for inputs raised to powers (page 53):

$$\frac{\delta [B^2]}{|B^2|} = 2 \frac{\delta B}{|B|} \qquad \frac{\delta [r^2]}{|r^2|} = 2 \frac{\delta r}{|r|}$$

Then apply the principle from page 61 to $\frac{e}{m} = \frac{2V_a}{B^2 r^2}$.

$$\frac{\delta [e/m]}{|e/m|} = \left(\left(\frac{\delta [B^2]}{|B^2|}\right)^2 + \left(\frac{\delta [r^2]}{|r^2|}\right)^2 + \left(\frac{\delta V_a}{|V_a|}\right)^2 \right)^{1/2}$$

$$= \left(\left(2 \frac{\delta B}{|B|}\right)^2 + \left(2 \frac{\delta r}{|r|}\right)^2 + \left(\frac{\delta V_a}{|V_a|}\right)^2 \right)^{1/2}$$

$$= \left(4\left(\left(\frac{\delta I}{|I|}\right)^2 + \left(\frac{\delta D}{|D|}\right)^2 \right) + 4\left(\frac{\delta r}{|r|}\right)^2 + \left(\frac{\delta V_a}{|V_a|}\right)^2 \right)^{1/2}$$

$$= \left(\left(\frac{\delta V_a}{V_a}\right)^2 + 4\left(\frac{\delta I}{I}\right)^2 + 4\left(\frac{\delta r}{r}\right)^2 + 4\left(\frac{\delta D}{D}\right)^2 \right)^{1/2}$$

(noting $V_a, I, r, D > 0$.)

Additional Questions

1. Before crossing the potential, the electron has potential energy eV_a. It finishes with a kinetic energy of $\frac{1}{2} m v^2$.

$$eV_a = \frac{1}{2} m v^2$$

$$\Rightarrow 2eV_a = mv^2 \Rightarrow v = \left(\frac{2eV_a}{m}\right)^{1/2}$$

$$v = \left(\frac{2 \cdot (1.6021\ldots \cdot 10^{-19} C) \cdot (2000\ V)}{(9.1093\ldots 10^{-31})} \right)^{1/2} \approx 3 \cdot 10^7 \frac{m}{s}$$

Using the uniform circular motion constraint and the magnitude of the Lorentz force on the electron:

$$m\frac{v^{\cancel{2}}}{r} = |-e\vec{v}\times\vec{B}| = e\cancel{v}B$$

$$\Rightarrow B = \frac{mv}{er} = \frac{(9.1093\ldots\cdot 10^{-31}\ \text{kg})(3\cdot 10^{7}\ \text{m/s})}{(1.6021\ldots\cdot 10^{-19}\ \text{C})(0.35\ \text{m})}$$

$$\approx 0.0005\ \text{T} = 0.5\ \text{mT}$$

$$= 500\ \mu\text{T}$$

2 $(\gamma-1)mc^2 = eV_a$

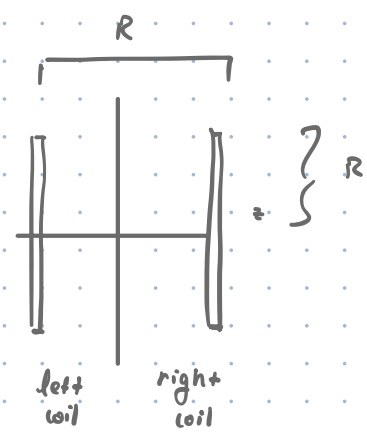
$$\gamma - 1 = \frac{eV_a}{mc^2} \Rightarrow \gamma = \frac{eV_a}{mc^2} + 1$$

For $V_a = 2000\text{V}$ and e, m, c from CODATA values:

$$\gamma = \frac{e(2000\text{V})}{mc^2} + 1 \approx 1.0039$$

This Lorentz factor isn't yet much greater than 1, so relativistic corrections are not strictly necessary.

3 Suppose two coils are placed symmetrically about the origin and parallel, with a spacing of R, their radius.



Magnetic field contributions from different contributors superpose. In a Helmholtz coil both field contributions are in the same direction, so the contributions superpose constructively.

The contribution from the left coil B_L equals the contribution from the right B_R.

Define $B_L = B_R = \vec{B}$.

$$\vec{B} = \frac{\mu_0 I R^2}{2(R^2 + R^2/4)^{3/2}}.$$

The total contribution is

$$B = 2\hat{B} = \frac{\mu_0 I R^2}{(R^2 + R^2/4)^{3/2}} = \frac{\mu_0 I R^2}{(5R^2/4)^{3/2}} = \frac{\mu_0 I R^2}{5\sqrt{5} \cdot R^3/8}$$

$$= \frac{8 \mu_0 I}{5\sqrt{5}\, R} = \frac{16 \mu_0 I}{\sqrt{125}\, D}$$

where D is the coil diameter.

With N turns in each coil, $\hat{B}$ grows by a factor of N relative to its single-loop value:

$$B = 2 \cdot N \cdot \hat{B} = \frac{16 \mu_0 I N}{\sqrt{125}\, D}$$

4 For $V_a = 2000V$ $I = 0.1238 \pm 0.0001$ A was used.

The coil diameter was 15.5 ± 0.1 cm.

$$B = \frac{16 \mu_0 (0.1238\ A)(320)}{\sqrt{125}\,(0.155\ m)} = 0.00045\ldots\ T$$

$$\delta B = B \left(\left(\frac{\delta I}{I}\right)^2 + \left(\frac{\delta D}{D}\right)^2 \right)^{1/2}$$

$$= B \cdot \left(\left(\frac{0.0001}{0.1238}\right)^2 + \left(\frac{0.001}{0.155}\right)^2 \right)^{1/2}$$

$$\approx 3 \cdot 10^{-6}\ T$$

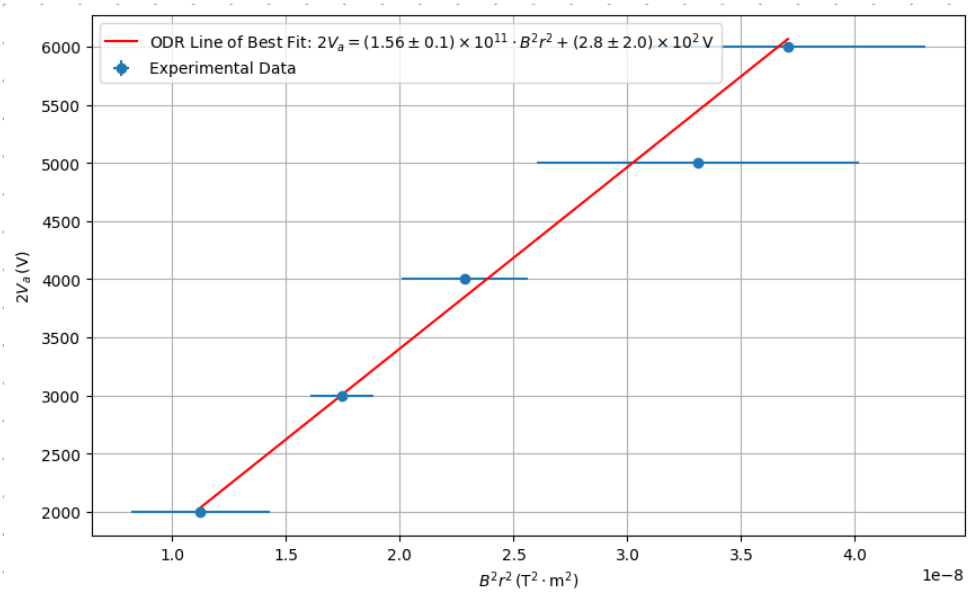
$$B = (459 \pm 3)\ \mu T$$

The value obtained from problem 1 was $500\ \mu T$, with is reasonably close to the estimated value of B for the 2000V trial.

The horizontal component of Earth's magnetic field is about $25\ \mu T$. This is not within the uncertainty computed above; Earth's field is likely to affect the experiment.

This procedure addressed this by computing y_{offset} values to correct y-coordinate readings with, obtained from a beam deflection corresponding to a 0 Helmholtz field. Subtracting this from y-coordinates aims to remove the influence of Earth's field.

This experiment also flipped the magnetic field and computed averages of y-coordinate magnitudes, reconciling for asymmetric deflection downward or upward.



$$\frac{e}{m} = \frac{2V_a}{B^2r^2} \Rightarrow 2V_a = \left(\frac{e}{m}\right) B^2r^2,$$

so $2V_a$ vs. B^2r^2 was plotted to more directly yield $\frac{e}{m}$ in the slope of the linear regression model.

This method yields $\frac{e}{m} = (1.56 \pm 0.1) \cdot 10^{11} \frac{C}{kg}$.